

Question Number	Answer	Additional Guidance	Mark
1(a)	A description that includes the following points: - dependent variable is the number (of eggs) hatched per unit time (1) - use of five different concentrations of calcium ions (1) - counting brine shrimps {hatched / swimming} at {same / stated} time intervals (1) - suitable control of one variable (1) - repeats (for each treatment) and calculate a mean (1)	Accept dependent variable is number (of eggs) hatched per stated time Accept use of zero calcium ions as one of the concentrations Ignore units if given If single count: minimum time one hour, maximum time 3 days eg. temperature / stated temp (up to 45°C) - thermostatically controlled waterbath pH - buffer light intensity – distance of bulb / wattage of bulb salinity - % or gml⁻¹	(5) exp

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	An answer that includes two of the following points: pH (1)		(2)

	• temperature (1) • salinity (1) • light intensity (1)		exp
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1(b)(ii)	An answer that includes the following points: - variable with suitable control method described (1) - results are not valid / description of expected effect on the dependent variable (1)	Description should be directional	(2) exp

Question Number	Answer	Additional Guidance	Mark
1(c)	An explanation that includes three of the following points: - calcium ions allow substrate to bind to the active site of enzyme (1) - activation energy lowered (at active site) (1) - an increased rate of respiration (1) - faster {anabolic / synthesis} reactions (leading to earlier hatching) (1)	Accept calcium ions increase binding of substrate to active site Accept calcium ions allow formation of enzyme- substrate complex Accept description of anabolic / synthesis reactions	(3) exp

(Total for question 1 = 12 marks)

